

LIST OF EXPERIMENTS

There are many different types of experiments that can be conducted. Here are some examples:

1. **List experiments:** These experiments present respondents with a list of items, experiences, or statements. Respondents must then report how many items in the list pertain to them. In a list experiment, the sample is randomly divided into two groups: the Direct Response Group and the Veiled Response Group. With a large enough sample size, list experiments can be used to estimate the proportion of people for whom a sensitive statement is true.
2. **Psychological experiments:** These experiments are designed to study human behavior and mental processes. Some famous examples include Harry Harlow's experiments with baby monkeys and wire and cloth surrogate mothers (1957–1974), Stanley Milgram's experiments on human obedience (1963), and Walter Mischel's marshmallow experiment showing the importance to life outcomes of the ability to delay gratification (beginning late 1960s).
3. **Science experiments:** These experiments are designed to explore scientific concepts and phenomena. There are many different types of science experiments that can be conducted, including kitchen chemistry experiments, DIY mini drones, and more. Science experiments can be conducted at home or in a school setting .

Note: A minimum of ten experiments from the following should be performed.

1. Experiment 1: Description of experiment 1.
2. Experiment 2: Description of experiment 2.
3. Experiment 3: Description of experiment 3.
4. Experiment 4: Description of experiment 4.
5. Experiment 5: Description of experiment 5.
6. Experiment 6: Description of experiment 6.
7. Experiment 7: Description of experiment 7.
8. Experiment 8: Description of experiment 8.
9. Experiment 9: Description of experiment 9.
10. Experiment 10: Description of experiment 10.

(A) Hardware based experiments

Hardware-based experiments are a type of scientific experiment that involves the use of physical equipment or machinery to test a hypothesis or observe a phenomenon. These experiments can be conducted in a variety of fields, including physics, chemistry, biology, and engineering.

Some examples of hardware-based experiments include:

1. Testing the strength of materials: In this type of experiment, a material is subjected to various forces to determine its strength and durability. This can be done using machines that apply pressure, tension, or torsion to the material.
2. Measuring the speed of sound: This experiment involves using specialized equipment to measure the speed at which sound travels through different mediums, such as air, water, or solids.
3. Observing chemical reactions: In this type of experiment, chemicals are mixed together and the resulting reaction is observed and recorded. This can be done using equipment such as beakers, test tubes, and microscopes.
4. Studying the behavior of particles: Particle accelerators and detectors can be used to study the behavior of subatomic particles, such as electrons and protons.

Hardware-based experiments can provide valuable data and insights into the natural world, and are an important tool in scientific research. However, they can also be expensive and time-consuming to conduct, and may require specialized equipment and expertise.

1. Verification of Kirchhoff's laws

Kirchhoff's laws are two equalities that deal with the current and potential difference (commonly known as voltage) in the lumped element model of electrical circuits. They were first described in 1845 by German physicist Gustav Kirchhoff. These laws are used to determine the values of the current and voltage in an electrical circuit.

- **Kirchhoff's Current Law (KCL):** This law is also known as Kirchhoff's first law, Kirchhoff's point rule, or Kirchhoff's junction rule. It states that, for any node (junction) in an electrical circuit, the sum of currents flowing into that node is equal to the sum of currents flowing out of that node. Mathematically, the algebraic sum of currents in a network of conductors meeting at a point is zero.
- **Kirchhoff's Voltage Law (KVL):** This law is also known as Kirchhoff's second law, Kirchhoff's loop (or mesh) rule, and Kirchhoff's second rule. It states that the directed sum of the potential differences (voltages) around any closed network is zero. Mathematically, the algebraic sum of the products of the resistances of the conductors and the currents in them in a closed loop is equal to the total emf available in that loop.

Verification of Kirchhoff's laws can be done by setting up a circuit and measuring the current and voltage at different points in the circuit. The measured values can then be compared to the values predicted by Kirchhoff's laws to verify their accuracy. This can be done using a multimeter to measure the current and voltage, and a known resistance to calculate the current using Ohm's law. The

experiment can be repeated with different circuit configurations to further verify the accuracy of Kirchhoff's laws.

2. Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor

- Power is the rate at which energy is transferred or used. In an AC circuit, power is measured in watts (W).
- Power factor is the ratio of real power to apparent power. It is a measure of how effectively the power is being used in a circuit.
- In a single phase AC series inductive circuit, the power factor is typically less than 1 due to the presence of inductive reactance.
- The power factor can be improved by adding a capacitor in parallel with the inductive load. This is known as power factor correction.
- Power factor correction reduces the reactive power drawn from the power source, which can result in reduced energy costs and improved system efficiency.
- To measure the power and power factor in a single phase AC series inductive circuit, a power meter or power analyzer can be used. These instruments measure the voltage, current, and power factor of the circuit.
- The power factor can also be calculated using the formula: power factor = real power / apparent power.
- By measuring the power and power factor in a single phase AC series inductive circuit, it is possible to determine the need for power factor correction and to select the appropriate capacitor value for improving the power factor.

3. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.

- Resonance is a phenomenon that occurs in RLC series circuits when the circuit's inductive reactance is equal to its capacitive reactance.
- At resonance, the circuit's impedance is purely resistive, and the current and voltage are in phase.
- The resonant frequency of an RLC series circuit is given by the formula: $f = 1 / (2 * \pi * \sqrt{L * C})$, where f is the resonant frequency, L is the inductance, and C is the capacitance.
- To obtain the resonant frequency of an RLC series circuit, the values of the circuit's resistance, inductance, and capacitance must be known.
- The resonant frequency can be determined experimentally by measuring the circuit's impedance at different frequencies and finding the frequency at which the impedance is purely resistive.
- At resonance, the circuit exhibits maximum power transfer and minimum impedance.
- The phenomenon of resonance has many practical applications, including

in the design of filters, oscillators, and tuned circuits.

4. Connection and measurement of power consumption of a fluorescent lamp (tube light)

A fluorescent lamp, also known as a tube light, is a low-pressure mercury-vapor gas-discharge lamp that uses fluorescence to produce visible light. To measure the power consumption of a fluorescent lamp, the following steps can be taken:

1. **Connect the lamp to a power source:** The fluorescent lamp should be connected to a power source using the appropriate wiring and connections. Ensure that the connections are secure and that the lamp is properly installed.
2. **Measure the current:** Use a clamp meter to measure the current flowing through the circuit. Place the clamp around one of the wires connected to the lamp and take the reading.
3. **Measure the voltage:** Use a multimeter to measure the voltage across the lamp. Place the probes of the multimeter on the appropriate points in the circuit and take the reading.
4. **Calculate the power consumption:** The power consumption of the lamp can be calculated by multiplying the current and voltage readings. The result will be the power consumption in watts.

It is important to note that the power consumption of a fluorescent lamp may vary depending on factors such as the age of the lamp, the temperature of the environment, and the quality of the power supply. Regularly measuring the power consumption of a fluorescent lamp can help to identify any changes in its performance and efficiency.

5. Measurement of power in 3- phase circuit by two-wattmeter method and determination of its power factor for star as well as delta connected load.

- The two-wattmeter method is used to measure the power in a three-phase circuit.
- This method involves the use of two wattmeters, which are connected to measure the power in two of the three phases.
- The power in the third phase is calculated by subtracting the sum of the readings of the two wattmeters from the total power.
- The power factor of the circuit can be determined by calculating the ratio of the total power to the product of the line voltage, line current, and the square root of three.
- This method can be used for both star and delta connected loads.
- The power factor is an important parameter in the analysis of three-phase circuits, as it indicates the efficiency of the circuit in converting electrical energy into useful work.

- A high power factor indicates that the circuit is operating efficiently, while a low power factor indicates that there is a significant amount of reactive power in the circuit, which reduces its efficiency.

6. Determination of parameters of ac single phase series RLC circuit

An RLC circuit is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series. The circuit is driven by an alternating current (AC) voltage source. The parameters of an RLC circuit can be determined by measuring the voltage and current at different points in the circuit and using the appropriate equations.

1. **Resistance (R):** The resistance of the circuit can be determined by measuring the voltage across the resistor and the current flowing through it. The resistance is calculated using Ohm's law, which states that the voltage across a resistor is equal to the product of the current flowing through it and the resistance: $V = IR$.
2. **Inductance (L):** The inductance of the circuit can be determined by measuring the voltage across the inductor and the current flowing through it. The inductance is calculated using the equation $V = L * (dI/dt)$, where dI/dt is the rate of change of current with respect to time.
3. **Capacitance (C):** The capacitance of the circuit can be determined by measuring the voltage across the capacitor and the current flowing through it. The capacitance is calculated using the equation $I = C * (dV/dt)$, where dV/dt is the rate of change of voltage with respect to time.
4. **Impedance (Z):** The impedance of the circuit is the total opposition to the flow of current. It can be calculated using the equation $Z = R + j(XL - XC)$, where XL is the inductive reactance, XC is the capacitive reactance, and j is the imaginary unit.
5. **Resonant frequency (f):** The resonant frequency of the circuit is the frequency at which the inductive reactance and the capacitive reactance are equal. It can be calculated using the equation $f = 1 / (2 * \pi * \text{sqrt}(L * C))$.

These are the main parameters that can be determined for an AC single phase series RLC circuit. By measuring the voltage and current at different points in the circuit and using the appropriate equations, the values of these parameters can be calculated.

7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer

A single-phase transformer is an electrical device that is used to transfer electrical energy from one circuit to another by means of electromagnetic induction.

The voltage ratio, polarity, and efficiency of a single-phase transformer can be determined by performing a load test.

- i. **Voltage Ratio:** The voltage ratio of a transformer is the ratio of the secondary voltage to the primary voltage. It can be determined by measuring the primary and secondary voltages of the transformer while it is under load and calculating the ratio.
- ii. **Polarity:** The polarity of a transformer refers to the direction of the induced voltage in the secondary winding relative to the primary winding. It can be determined by performing a polarity test, which involves connecting the primary and secondary windings in series and measuring the resulting voltage.
- iii. **Efficiency:** The efficiency of a transformer is the ratio of the output power to the input power. It can be determined by measuring the input and output power of the transformer while it is under load and calculating the ratio.

During a load test, the transformer is connected to a load and the primary and secondary voltages, as well as the input and output power, are measured. These measurements can then be used to calculate the voltage ratio, polarity, and efficiency of the transformer. It is important to note that the load test should be performed under normal operating conditions to ensure accurate results.

8. Determination of efficiency of a dc shunt motor by load test

The efficiency of a DC shunt motor can be determined by performing a load test. The load test involves measuring the input power, output power, and losses of the motor at different load conditions. The steps to determine the efficiency of a DC shunt motor by load test are as follows:

1. Connect the DC shunt motor to a suitable load and power supply.
2. Measure the input voltage and current to the motor using a voltmeter and ammeter, respectively.
3. Measure the speed of the motor using a tachometer.
4. Calculate the input power to the motor by multiplying the input voltage and current.
5. Calculate the output power of the motor by multiplying the torque and speed of the motor.
6. Calculate the losses in the motor by subtracting the output power from the input power.
7. Calculate the efficiency of the motor by dividing the output power by the input power and multiplying by 100%.

The efficiency of the motor can be plotted against the load to determine the optimal operating conditions for the motor. It is important to note that the efficiency of a DC shunt motor will vary with the load and operating conditions.

Therefore, it is important to perform the load test at different load conditions to accurately determine the efficiency of the motor.

9. To study running and speed reversal of a three phase induction motor and record speed in both directions.

- A three-phase induction motor is an AC electric motor that converts electrical energy into mechanical energy.
- The motor consists of a stator, which is a stationary part, and a rotor, which is a rotating part.
- The stator has a three-phase winding, which produces a rotating magnetic field when a three-phase AC supply is applied.
- The rotating magnetic field induces an electromotive force in the rotor, which causes it to rotate.
- The speed of the motor can be controlled by varying the frequency of the AC supply.
- To reverse the direction of rotation of the motor, the phase sequence of the AC supply must be changed.
- This can be achieved by interchanging any two of the three-phase supply connections to the motor.
- The speed of the motor in both directions can be recorded using a tachometer or a stroboscope.

10. Demonstration of cut-out sections of machines: dc machine, three phase induction machine, single-phase induction machine and synchronous machine.

- **DC Machine:** A DC machine is an electromechanical device that converts electrical energy into mechanical energy or vice versa. The cut-out section of a DC machine shows the armature, commutator, brushes, and field windings.
- **Three Phase Induction Machine:** A three-phase induction machine is an AC motor that converts electrical energy into mechanical energy. The cut-out section of a three-phase induction machine shows the stator, rotor, and windings.
- **Single-Phase Induction Machine:** A single-phase induction machine is an AC motor that operates on a single-phase power supply. The cut-out section of a single-phase induction machine shows the stator, rotor, and windings.
- **Synchronous Machine:** A synchronous machine is an AC machine that operates at a constant speed, determined by the frequency of the power supply. The cut-out section of a synchronous machine shows the stator, rotor, and field windings.

These cut-out sections provide a visual representation of the internal components and workings of these machines, allowing for a better understanding of their operation and maintenance.

(B) Experiments available on virtual lab

There are a variety of virtual lab experiments available online for students to explore and learn from. Some of the virtual lab experiments available are:

1. **Virtual Chemistry Experiments** - A collection of interactive web-based chemistry tutorials .
2. **Mixed Reception** - Solve a mysterious death by gathering clues, interviewing suspects, and analyzing evidence .
3. **The Virtual Lab** - An online simulation of a chemistry lab designed to help students link chemical computations with authentic laboratory chemistry .
4. **PraxiLabs** - Offers virtual experiments that adapt to any learning style, with progress tracking, walkthrough videos, lab manuals, enrichment documents, immediate feedback, MCQ quizzes, and much more .
5. **VRLab Academy** - Offers more than 200 science experiments available on VR and PC, with virtual reality technology to increase the ability to learn through virtual memory .
6. **American Chemical Society** - Offers virtual chemistry and simulations, including the chemistry of aspirin, with four levels of experiments, including synthesis, thin layer chromatography, and reaction conditions .
7. **Mystery Science** - Offers a wealth of virtual science labs, activities, and experiments for students in elementary school, covering topics from weather to astronomy, humans to animals .
8. **Molecular Workbench** - Offers hundreds of virtual science labs for physics, biology, chemistry, and more .
9. **National Science Digital Library** - Offers ChemCollective Virtual Labs, a deep dive into chemistry with virtual labs .
10. **PBS Nova Labs** - A platform for teens and older learners to explore and examine cutting-edge issues .

These are just a few examples of the many virtual lab experiments available online for students to explore and learn from.

1. Kirchhoff's laws

Kirchhoff's laws are two laws that deal with the conservation of charge and energy in electrical circuits. These laws are named after the German physicist Gustav Kirchhoff.

1. Kirchhoff's Current Law (KCL): This law states that the algebraic sum of currents entering a node (or a closed boundary) is zero. In other words, the total current entering a node must be equal to the total current leaving

the node. This law is based on the principle of conservation of electric charge.

2. Kirchhoff's Voltage Law (KVL): This law states that the algebraic sum of the potential differences (voltages) around any closed loop is zero. In other words, the total voltage gained in a closed loop must be equal to the total voltage lost. This law is based on the principle of conservation of energy.

These laws are widely used in circuit analysis to determine the current and voltage in different parts of a circuit. They are fundamental principles that must be satisfied in any circuit, regardless of its complexity.

Virtual Lab

Virtual Lab at Amrita uses state-of-the-art computer simulation technology to create real world environments and problem handling capabilities. It is required to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics, Chemistry and Biotechnology .

Virtual labs are media-rich online learning environments, which allow users to conduct physical laboratory experiments in a computer-simulated environment. This innovative new educational technique is enabling experimentation beyond the boundaries of traditional laboratories .

The link you provided is for the Kirchhoff's Laws simulation in the Amrita Vishwa Vidyapeetham Virtual Lab. This simulation allows users to learn about Kirchhoff's Voltage Law, which states that "The algebraic sum of all voltages in a loop must equal zero" . The simulator includes components such as resistors, which produce a voltage across its terminals that is proportional to the electric current flowing through it in accordance with Ohm's law .

2. Thevenin Theorem

Thevenin's Theorem is a technique used to analyze complex electrical circuits by simplifying them into an equivalent circuit composed of a single voltage source, a single resistor, and a load. The theorem states that any linear, bilateral circuit can be replaced by an equivalent circuit consisting of a voltage source and a series resistor.

The steps to apply Thevenin's Theorem are as follows: 1. Identify the load resistor in the circuit, which is the resistor across which the voltage or current is to be determined. 2. Remove the load resistor from the circuit. 3. Calculate the open-circuit voltage across the terminals where the load resistor was connected. This is the Thevenin voltage. 4. Calculate the equivalent resistance of the circuit as seen from the open terminals, with all voltage sources replaced by short circuits and all current sources replaced by open circuits. This is the Thevenin resistance. 5. Replace the original circuit with the Thevenin equivalent circuit,

consisting of the Thevenin voltage source in series with the Thevenin resistance and the load resistor.

Thevenin's Theorem can be used to analyze both DC and AC circuits. It is a powerful tool for simplifying complex circuits and making circuit analysis more manageable.

Virtual Lab

Virtual Lab is an initiative by Amrita University, Dayalbagh University, NIT Karnataka, and COE Pune. The main objective of Virtual Labs is to provide remote access to labs in various disciplines of Science and Engineering. These Virtual Labs cater to students at the undergraduate level, postgraduate level, and research scholars .

Amrita Virtual Biotechnology laboratory is a new kind of experimental science that is being established as a virtual simulation-based laboratory. These labs focus on helping students retain the real feel of a laboratory while conducting the experiment from an internet-enabled computer terminal, much in the same way as is done in a real lab .

The VALUE Initiative is a project dedicated to developing virtual labs in various science and engineering disciplines. Virtual Lab at Amrita uses state-of-the-art computer simulation technology to create real-world environments and problem-handling capabilities. This is required to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics .

Optics Virtual Lab is one of the labs available under the Physical Sciences category. Optics is the study of the behavior and properties of light, including its interactions with matter and the construction of instruments that use or detect it. Optics usually describes the behavior of visible, ultraviolet, and infrared light .

The Online Labs is based on the idea that lab experiments can be taught using the Internet, more efficiently, and less expensively. The labs can also be made available to students with no access to physical labs or where equipment is not available owing to being scarce or costly .

3. RLC Series Resonance

- An RLC circuit is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series.
- At resonance, the impedance of the circuit is at its minimum, and the current flowing through the circuit is at its maximum.
- The resonant frequency of the circuit is given by the formula: $f = 1 / (2 * \pi * \sqrt{L * C})$.
- The quality factor (Q) of the circuit is given by the formula: $Q = (1 / R) * \sqrt{L / C}$.

- The bandwidth of the circuit is given by the formula: $BW = f / Q$.
- The voltage across the resistor, inductor, and capacitor are in phase at resonance.
- The voltage across the inductor and capacitor are equal in magnitude but opposite in phase at resonance.
- The power factor of the circuit is 1 at resonance.
- The circuit behaves as a band-pass filter at resonance, allowing only the frequencies near the resonant frequency to pass through.

Virtual Lab

Virtual labs are media-rich online learning environments, which allow users to conduct physical laboratory experiments in a computer-simulated environment. This innovative new educational technique is enabling experimentation beyond the boundaries of traditional laboratories .

- These Virtual Labs cater to students at the undergraduate level, postgraduate level, as well as to research scholars .
- The aim is to enthuse students to conduct experiments by arousing their curiosity. This would help them in learning basic and advanced concepts through remote experimentation .
- This project is an initiative of the Ministry of Human Resource Department under the National Mission on Education through ICT .
- These experiments and labs are hosted for open access through the main project website .
- Virtual Lab at Amrita uses state-of-the-art computer simulation technology to create real-world environments and problem-handling capabilities .
- The goal is to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics .
- Amrita CAP-VL provides a platform for the end-to-end management of such distributed labs so that access to labs by various players can be brought under one umbrella .
- Each lab may be developed using different technologies based on the lab developer preferences .

4. Measurement of power in 3- phase circuit by two wattmeter method and determination of its power factor for star as well as delta connected load.

- The two wattmeter method is used to measure the total power in a three-phase circuit.
- This method is applicable to both balanced and unbalanced three-phase loads.
- Two wattmeters are connected to the circuit, with one wattmeter measuring the power in one phase and the other wattmeter measuring the power

in another phase.

- The sum of the readings of the two wattmeters gives the total power in the circuit.
- The power factor of the circuit can be determined by the phase angle between the voltage and current of the circuit.
- For a star-connected load, the power factor is given by the cosine of the phase angle between the line voltage and line current.
- For a delta-connected load, the power factor is given by the cosine of the phase angle between the phase voltage and phase current.
- The two wattmeter method is a simple and accurate way to measure the power and power factor in a three-phase circuit.

Virtual Lab

Virtual Lab is an online platform that provides students with the opportunity to conduct experiments and learn about various scientific concepts in a virtual environment. The link you provided is for the Virtual Lab at the Dayalbagh Educational Institute, which focuses on the topic of measurement using Dreamweaver.

Some key points to note about this Virtual Lab are: - It provides an interactive and engaging learning experience for students. - The experiments are designed to help students understand the concepts of measurement and data analysis. - The virtual environment allows students to conduct experiments without the need for physical equipment or materials. - The platform is accessible from anywhere with an internet connection, making it a convenient learning tool for students.

5. Determination of parameters of ac single phase series RLC circuit

An AC single-phase series RLC circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series with an AC voltage source. The parameters of this circuit can be determined using the following steps:

1. **Determine the impedance (Z):** The impedance of the circuit is the total opposition to the flow of current. It is calculated using the formula $Z = \sqrt{R^2 + (X_L - X_C)^2}$, where X_L is the inductive reactance and X_C is the capacitive reactance.
2. **Determine the current (I):** The current flowing through the circuit can be calculated using Ohm's law, $I = V/Z$, where V is the applied voltage.
3. **Determine the voltage across each component:** The voltage across each component can be calculated using Ohm's law, $V_R = IR$, $V_L = IL$, and $V_C = IC$, where V_R , V_L , and V_C are the voltages across the resistor, inductor, and capacitor, respectively.

4. **Determine the power factor:** The power factor is the ratio of the real power to the apparent power. It can be calculated using the formula $\text{PF} = \cos(\Phi)$, where Φ is the phase angle between the voltage and current.
5. **Determine the resonance frequency:** The resonance frequency is the frequency at which the inductive reactance and capacitive reactance are equal. It can be calculated using the formula $f = 1/(2\sqrt{LC})$.

These are the basic steps to determine the parameters of an AC single-phase series RLC circuit. It is important to note that the values of the parameters may vary depending on the frequency of the applied voltage.

Virtual Lab

- Virtual Labs is a project initiated by the Ministry of Education, Government of India.
- It falls under the National Mission on Education through Information and Communication Technology.
- The project aims to provide remote access to Laboratories in various disciplines of Science and Engineering.
- It is for students at all levels from undergraduate to research.

6. To observe the B-H loop of a ferromagnetic material in CRO

- A B-H loop, also known as a hysteresis loop, is a graphical representation of the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a ferromagnetic material.
- The B-H loop can be observed using a Cathode Ray Oscilloscope (CRO).
- To observe the B-H loop, a ferromagnetic material is subjected to an alternating magnetic field.
- As the magnetic field strength is increased, the magnetic flux density of the material also increases, until it reaches a maximum value called the saturation magnetization.
- When the magnetic field strength is decreased, the magnetic flux density does not return to zero, but instead, it decreases to a value called the remanent magnetization.
- This phenomenon is known as hysteresis, and it is the reason why the B-H loop has a characteristic shape.
- The area enclosed by the B-H loop is a measure of the energy lost in the material due to hysteresis.
- The B-H loop can provide valuable information about the magnetic properties of a ferromagnetic material, such as its coercivity, remanent magnetization, and saturation magnetization.

Virtual Lab

Virtual Lab is an initiative of the Ministry of Human Resource Department under the National Mission on Education through ICT. The project aims to

provide remote access to labs in various disciplines of Science and Engineering. These Virtual Labs cater to students at the undergraduate level, postgraduate level, as well as to research scholars .

One of the Virtual Labs is hosted by Amrita Vishwa Vidyapeetham and uses state-of-the-art computer simulation technology to create real-world environments and problem-handling capabilities. This is done to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics .

One of the experiments available in the Virtual Lab is the Zener Diode as Voltage Regulator experiment. This is an ideal circuit, and for a real circuit, there will be small variations in output voltage with varying input voltage or load resistance .

Another experiment available in the Virtual Lab is the Solid State Physics Virtual Lab. This remote trigger experiment performs experiments on actual hysteresis loop tracer equipment (HLT-111). The experimental setup is housed at Amrita University, Amritapuri Campus .

7. Determination of the efficiency of a dc motor by loss summation method (Swinburne's test)

- Swinburne's test is the simplest indirect method of testing dc machines.
- In this method, the dc machine (generator or motor) is run as a motor at no-load and losses of the machine are determined.
- Once the losses of the machine are known, its efficiency at any desired load can be determined in advance.
- Swinburne's test is an indirect method of testing DC machines, named after Sir James Swinburne.
- The Swinburne's test is the simplest method of testing of shunt and compound DC machines which have constant field flux.
- The losses in the motors are armature copper losses, iron losses in the core, friction losses, and winding losses.
- These losses are calculated separately and efficiency can be pre-determined.

Course Outcomes:

- Understand the fundamental concepts and principles of the subject.
- Develop the ability to apply the knowledge and skills acquired in the course to solve problems.
- Enhance critical thinking and analytical skills.
- Improve communication skills through written and oral presentations.
- Develop the ability to work effectively in a team.
- Foster a sense of curiosity and a desire for lifelong learning.
- Prepare for further study or professional development in the field.

Course Outcome (CO) Bloom's Level

Course Outcome (CO) refers to the expected knowledge, skills, and attitudes that students should possess upon completion of a specific course. These outcomes are used to assess the effectiveness of the course in achieving its intended goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is commonly used to design assessments and learning activities that align with the desired course outcomes.

The levels of Bloom's Taxonomy, in ascending order of complexity, are:

1. Remembering: recalling previously learned information.
2. Understanding: comprehending the meaning of the information.
3. Applying: using the information in a new situation.
4. Analyzing: breaking down the information into parts to understand its structure.
5. Evaluating: making judgments about the value of the information.
6. Creating: combining information to form a new whole.

By mapping Course Outcomes to the levels of Bloom's Taxonomy, educators can ensure that their assessments and learning activities are appropriately challenging and aligned with the desired learning outcomes. This can help students achieve a deeper understanding of the course material and develop the skills and knowledge necessary for success in their field of study.

At the end of this course, the students should be able to:

1. Understand the fundamental concepts and principles of the subject.
2. Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
3. Communicate effectively using the language and terminology of the subject.
4. Work collaboratively with others to achieve common goals.
5. Demonstrate critical thinking and analytical skills.
6. Develop a lifelong interest in learning and self-improvement.
7. Demonstrate ethical and responsible behavior.
8. Use technology effectively to enhance learning and productivity.
9. Appreciate the relevance and importance of the subject in the real world.
10. Demonstrate mastery of the subject matter through assessments and practical applications.

CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3

- KVL (Kirchhoff's Voltage Law) states that the sum of the voltage drops around a closed loop in a circuit is equal to the voltage supplied to the

loop.

- KCL (Kirchhoff's Current Law) states that the sum of the currents entering a node in a circuit is equal to the sum of the currents leaving the node.
- Network theorems, such as Thevenin's theorem and Norton's theorem, can be used to simplify complex circuits and make it easier to analyze them.
- To conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits, one can set up a simple circuit with a power source, resistors, and measuring devices such as a voltmeter and an ammeter.
- By measuring the voltage drops across the resistors and the currents flowing through them, one can verify the validity of KVL and KCL.
- By applying network theorems, one can simplify the circuit and make predictions about its behavior, which can then be verified through experimentation.

CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measure power in single phase as well as three phase electrical circuits. K4

- An AC circuit is an electrical circuit that is powered by an alternating current (AC) source.
- A single-phase AC supply is an electric power supply that uses a single sinusoidal voltage waveform.
- The behavior of an AC circuit connected to a single-phase AC supply can be demonstrated by analyzing the voltage, current, and power in the circuit.
- The power in a single-phase AC circuit can be measured using a wattmeter.
- A three-phase electrical circuit is a type of polyphase system that uses three conductors carrying alternating currents of the same frequency and amplitude, but with a phase difference of 120 degrees between them.
- The power in a three-phase electrical circuit can be measured using a three-phase power meter or by measuring the power in each phase and summing the results.
- It is important to understand the behavior of AC circuits connected to single-phase and three-phase AC supplies in order to design and operate electrical systems effectively.

CO 3 Perform experiment illustrating BH curve of magnetic materials. K3

The BH curve, also known as the hysteresis loop, is a graphical representation of the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a magnetic material. The curve is used to illustrate the magnetic properties of a material, including its magnetization, saturation, and coercivity.

To perform an experiment illustrating the BH curve of magnetic materials, the

following steps can be followed:

1. Prepare a sample of the magnetic material to be tested. This can be in the form of a ring or a toroid.
2. Set up the test equipment, including a magnetometer, a power supply, and an oscilloscope.
3. Apply an alternating current to the sample to create a magnetic field.
4. Measure the magnetic flux density (B) of the sample using the magnetometer.
5. Plot the magnetic field strength (H) against the magnetic flux density (B) to create the BH curve.
6. Observe the shape of the curve and analyze the magnetic properties of the material, including its magnetization, saturation, and coercivity.

This experiment can be used to illustrate the magnetic properties of different materials and to compare their behavior under different conditions. It is an important tool in the study of magnetism and magnetic materials.

CO 4 Calculate efficiency of a single phase transformer and DC machine. K4

Efficiency is the ratio of output power to input power. It is a measure of how well a device converts the power it receives into useful work. The efficiency of a single phase transformer and a DC machine can be calculated using the following steps:

1. **Determine the input power:** Measure the voltage and current at the input of the device and use the formula $P = VI$ to calculate the input power.
2. **Determine the output power:** Measure the voltage and current at the output of the device and use the formula $P = VI$ to calculate the output power.
3. **Calculate the efficiency:** Divide the output power by the input power and multiply by 100 to express the efficiency as a percentage. The formula for efficiency is $\text{Efficiency} = (\text{Output Power} / \text{Input Power}) * 100$.

For a single phase transformer, the input power is the power supplied to the primary winding and the output power is the power delivered by the secondary winding. For a DC machine, the input power is the electrical power supplied to the armature and the output power is the mechanical power delivered by the shaft.

It is important to note that the efficiency of a device is not constant and can vary depending on the operating conditions. Factors such as load, temperature, and frequency can affect the efficiency of a single phase transformer and a DC machine. It is therefore important to measure the efficiency under the specific operating conditions for which the device is intended to be used.

CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4

1. **Speed measurement of three phase induction motor:** The speed of a three phase induction motor can be measured using a tachometer, stroboscope, or by counting the number of revolutions per minute (RPM) using a stopwatch. The synchronous speed of the motor can be calculated using the formula: $N_s = 120f/P$, where N_s is the synchronous speed, f is the frequency of the supply, and P is the number of poles.
2. **Reversal of direction of three phase induction motor:** The direction of rotation of a three phase induction motor can be reversed by interchanging any two of the three supply lines. This changes the direction of the rotating magnetic field, causing the rotor to rotate in the opposite direction.
3. **Identification of DC and AC machines based on their construction:** DC machines can be identified by the presence of a commutator, which is used to convert the alternating current in the armature windings to direct current. AC machines, on the other hand, do not have a commutator and rely on the alternating current supplied to the stator windings to produce a rotating magnetic field. The rotor of an AC machine can be of the squirrel cage or wound type, while the rotor of a DC machine is typically of the wound type.

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

These are the six levels of Bloom's Taxonomy, a framework for categorizing educational goals and objectives.

1. **K1 – Remember:** This level involves recalling information from memory. It includes recognizing and recalling facts, terms, basic concepts, and answers.
2. **K2 – Understand:** This level involves understanding the meaning of information. It includes interpreting, exemplifying, classifying, summarizing, inferring, comparing, and explaining.
3. **K3 – Apply:** This level involves using information in a new situation. It includes executing and implementing.
4. **K4 – Analyze:** This level involves breaking information into parts to understand its structure. It includes differentiating, organizing, and attributing.
5. **K5 – Evaluate:** This level involves making judgments based on criteria and standards. It includes checking and critiquing.
6. **K6 – Create:** This level involves putting elements together to form a new whole. It includes generating, planning, and producing.

These levels can be used to design educational activities and assessments, and to evaluate the effectiveness of educational programs. They can also be used by learners to guide their own learning and to assess their own progress.